# DevDocs — Citation & Source Display

## 1. Overview

DevDocs provides citations for project-specific answers so users can identify where the information came from.

Citations are generated by the **application layer using retrieved chunk metadata**. The LLM does not generate, select, or format citations.

The citation system is designed to:

- Keep answers clean and readable
- Show the source files used by the RAG pipeline
- Provide precise line references for code
- Allow users to open the referenced file directly
- Preserve citation behavior for older conversations

---

## 2. Citation Location

Citations are displayed **at the end of the answer**.

Example:

```
The authentication middleware validates the JWT before allowing
the request to continue.

Sources
- src/auth.py — Lines 10–20, 35–48
- src/jwt.py — Lines 15–29
```

Citations should not be inserted inline within the answer text.

---

## 3. Citation Format

DevDocs uses a simple text-based citation format.

No rich source cards or additional source metadata are required for V1.

### Code files

```
src/auth.py — Lines 10–20
```

### Non-code files

```
docs/architecture.md
```

Non-code files do not display line numbers.

---

## 4. Source Metadata

Code chunks contain line-range metadata:

```
line_start
line_end
```

Example:

```
{
    "file_path": "src/auth.py",
    "line_start": 10,
    "line_end": 20
}
```

This metadata is used by the application to construct the citation.

Line numbers are not generated or inferred by the LLM.

---

## 5. Citation Generation

Citation generation is handled by the **application layer after retrieval**.

The LLM receives the retrieved chunks and generates the answer normally.

The application already has access to the metadata of those retrieved chunks, so no additional LLM call is required to generate citations.

### Flow

```
User Question
      ↓
RAG Retrieval
      ↓
Retrieved Chunks + Metadata
      ↓
      ├──────────────→ LLM → Answer
```

```
        │
        └──────────────→ Application → Citation Data
                                          ↓
                                  Format Sources
                                          ↓
                                  Answer + Sources
```

The LLM does not need to:

- Identify which chunks it used
- Generate citation IDs
- Generate file paths
- Generate line numbers
- Format the Sources section

---

## 6. Source Selection

Sources are generated from the **final retrieved chunks** used by the RAG pipeline.

The application does not ask the LLM to determine which chunks it used.

This avoids unnecessary LLM reasoning, additional latency, and additional implementation complexity.

---

## 7. Multiple Chunks from the Same File

If multiple retrieved chunks belong to the same file, their line ranges are combined into a single citation for that file.

Example:

```
Chunk 1 → src/auth.py → Lines 10–20
Chunk 2 → src/auth.py → Lines 35–48
Chunk 3 → src/auth.py → Lines 70–82
```

The displayed citation becomes:

```
src/auth.py — Lines 10–20, 35–48, 70–82
```

The ranges are **not merged or modified**.

For example:

```
10–20
21–30
```

remains:

```
10–20, 21–30
```

The application simply preserves the individual retrieved ranges.

---

## 8. Source Ordering

Sources maintain the **retrieval order**.

The application should not alphabetically sort or otherwise reorder sources.

Example retrieval order:

```
1. src/auth.py
2. src/jwt.py
3. docs/security.md
```

The Sources section maintains that order.

When multiple chunks from the same file are combined, their line ranges also follow their retrieval order.

---

## 9. Duplicate Sources

Exact duplicate source references should not be displayed multiple times.

However, different line ranges from the same file are preserved as separate ranges within the same citation.

Example:

```
src/auth.py — Lines 10–20, 35–48
```

rather than:

```
src/auth.py — Lines 10–20
src/auth.py — Lines 35–48
```

---

## 10. Citation Click Behavior

Each citation is clickable.

When the user clicks a citation, DevDocs opens the referenced file in a **floating file viewer**.

The floating viewer should:

- Be large enough for comfortable reading
- Be smaller than the main application window
- Keep the current conversation visible behind it
- Use the existing internal file viewer
- Provide a clear `×` close/exit button

The user should not be navigated away from the current conversation.

---

## 11. Code Citation Navigation

When a code citation is clicked, the file viewer should:

1. Open the referenced file.
2. Display line numbers.
3. Automatically scroll to the cited line range.
4. Highlight the cited line range.

Example:

```
src/auth.py — Lines 35–48
```

The viewer opens `src/auth.py`, scrolls to the relevant area, and highlights lines 35–48.

---

## 12. Multiple Highlighted Ranges

When a citation contains multiple line ranges:

```
src/auth.py — Lines 10–20, 35–48, 70–82
```

the viewer should:

- Open `src/auth.py`
- Highlight all cited ranges
- Initially scroll to the first cited range

Example:

```
Lines 10–20   → highlighted
Lines 35–48   → highlighted
Lines 70–82   → highlighted
```

The ranges are not merged.

---

## 13. Non-Code Citation Behavior

Non-code files do not require line-number navigation.

When a non-code citation is clicked, the file simply opens in the floating file viewer.

Examples:

```
README.md
docs/architecture.pdf
docs/design.docx
config/settings.yaml
```

The appropriate existing file renderer should be used.

---

## 14. Deleted Files and Older Conversations

Conversation history is preserved even when the project changes.

An older assistant answer may contain a citation to a file that no longer exists in the current project.

Example:

```
Older answer:
Sources
- src/auth.py — Lines 10–20
```

If `src/auth.py` is later deleted and the user clicks the citation, DevDocs should display:

> **This file doesn't exist anymore.**

DevDocs should not attempt to retrieve or reconstruct the deleted file.

No RAG lookup is required for this case.

---

## 15. File Reprocessing

If a cited file is currently being reprocessed, no special citation state is required for V1.

The citation continues to reference the file.

If the file is still available, the file viewer can open the currently available file content.

If the file has been deleted, the deleted-file behavior applies:

> **This file doesn't exist anymore.**

---

## 16. Citation Persistence

Citations are part of the generated assistant response.

The application should preserve the citation information with the assistant message so that historical answers can continue to display their original sources.

The current project state may change after the answer was generated.

Therefore:

- The answer remains unchanged.
- The original citation remains visible.
- Clicking the citation checks whether the referenced file still exists.
- If it exists, open it.
- If it no longer exists, show the deleted-file message.

---

## 17. Mandatory Citations

Citations are mandatory for successful project-context answers.

Every successfully generated answer that uses project context must include a **Sources** section.

This applies even when the answer is very short.

Example:

```
Yes. Authentication is handled by the JWT middleware.

Sources
- src/auth/middleware.py — Lines 12–35
```

There is no separate rule for hiding citations on short answers.

## 18. Relationship with RAG

The citation system operates on the output of the retrieval pipeline.

The finalized retrieval flow is:

```
User Question
      ↓
Project + READY Filters
      ↓
BM25 Top-10
      +
Vector Top-10
      ↓
RRF
      ↓
Reranker
      ↓
Relevance Check
      ↓
Final Retrieved Chunks (≤ 5)
      ↓
LLM Answer Generation
      ↓
Application Citation Generation
      ↓
Sources Section
```

The citation system does not perform another retrieval operation.

---

## 19. Relationship with Streaming

DevDocs uses streaming for LLM responses.

The LLM answer can stream normally without citation information being generated by the LLM.

Conceptually:

```
LLM
 ↓
Streaming Answer
 ↓
Complete Answer
 ↓
Application generates Sources
```

```
  ↓
Sources appended/displayed
```

The citation metadata is already available from the retrieved chunks.

No additional LLM call is required.

---

## 20. Core V1 Principles

1. **Citations appear at the end of answers.**
2. **Citations are simple clickable text.**
3. **Code citations include line ranges.**
4. **Non-code citations do not include line numbers.**
5. **Line numbers come directly from chunk metadata.**
6. **Citation generation is handled by the application, not the LLM.**
7. **Multiple ranges from the same file are combined into one citation.**
8. **Individual line ranges are not merged.**
9. **Source ordering follows retrieval order.**
10. **Duplicate source references are removed.**
11. **Clicking a citation opens the file in a floating viewer.**
12. **Code citations scroll to and highlight the cited ranges.**
13. **The viewer has a clear close/exit button.**
14. **Deleted files from older conversations show “This file doesn't exist anymore.”**
15. **Every successful project-context answer includes citations.**

---

## 21. V1 Out of Scope

The following are not required for V1:

- LLM-generated citations
- LLM-based citation selection
- Inline citations
- Rich source cards
- Automatic citation ranking
- Citation confidence scores
- External web citations
- Citation verification using another LLM
- Automatic merging of adjacent/overlapping line ranges
- Reconstructing deleted files